Advanced glycation end product level, diabetes, and accelerated cognitive aging

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Objective: Several studies report that diabetes increases risk of cognitive impairment; some have hypothesized that advanced glycation end products (AGEs) underlie this association. AGEs are cross-linked products that result from reactions between glucose and proteins. Little is known about the association between peripheral AGE concentration and cognitive aging. Methods: We prospectively studied 920 elders without dementia, 495 with diabetes and 425 with normal glucose (mean age 74.0 years). Using mixed models, we examined baseline AGE concentration, measured with urine pentosidine and analyzed as tertile, and performance on the Modified Mini-Mental State Examination (3MS) and Digit Symbol Substitution Test (DSST) at baseline and repeatedly over 9 years. Incident cognitive impairment (a decline of >1.0 SD on each test) was analyzed with logistic regression. Results: Older adults with high pentosidine level had worse baseline DSST score (p=0.05) but not different 3MS score (p=0.32). On both tests, there was a more pronounced 9-year decline in those with high and mid pentosidine level compared to those in the lowest tertile (3MS 7.0, 5.4, and 2.5 point decline, p overall <0.001; DSST 5.9, 7.4, and 4.5 point decline, p=0.03). Incident cognitive impairment was higher in those with high or mid pentosidine level than those in the lowest tertile (3MS: 24% vs 17%, odds ratio=1.55; 95% confidence interval 1.07–2.26; DSST: 31% vs 22%, odds ratio=1.62; 95% confidence interval 1.13–2.33). There was no interaction between pentosidine level, diabetes status, and cognitive decline. Multivariate adjustment for age, sex, race, education, hypertension, cardiovascular disease, estimated glomerular filtration rate, and diabetes diminished results somewhat but overall patterns remained similar. Conclusion: High peripheral AGE level is associated with greater cognitive decline in older adults with and without diabetes.